

P15 — SecureChat: Web-based Secure Messaging App

1) Group number & project title

Group: P15

Project title: SecureChat — Web-based Secure Messaging App (AES-256-GCM + RSA-PSS / RSA-OAEP)

2) Group members

- Member 1: *Ali Asiri(202027780)* — 4
- Member 2: *Omar alshahrani(202040640)* — 4

3) Project objectives

- Implement end-to-end confidentiality using AES-256-GCM (per-message IVs) with session keys protected by RSA-OAEP.
- Provide message integrity & authentication via RSA-PSS signatures.
- Deliver a simple web UI, including key management and an Attack Lab to demonstrate the required attacks and defenses.

4) Plan & deliverables

We will submit the following final deliverables:

- **Source code repository** (complete project code).
- **Technical report** — 12–15 pages documenting design, threat model, attacks & defenses, and test results.
- **User guide** — 2–3 pages with setup and usage screenshots.
- **Demo video** — 5–7 minutes demonstrating normal operation, each attack, and the corresponding mitigation.
- **Submission bundle** — zipped repository including code, report, user guide, and demo video.